

# YIN Haoteng 殷浩騰

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## EDUCATION

- **Purdue University** West Lafayette, IN  
• **Ph.D. in Machine Learning**; Department of Computer Science 2019 – Current
- **Peking University** Beijing, P.R. China  
• **MPhil in Data Science**; Academy for Advanced Interdisciplinary Studies 2016 – 2019
- **Shandong University** Ji'nan, P.R. China  
• **BEng in Computer Science and Technology**; Taishan College\* (Honors School) 2012 – 2016  
• **BEcon in Finance**; School of Economics 2014 – 2016  
\* Top-notch Undergraduate Training Program by Ministry of Education, P.R. China

## RESEARCH INTERESTS

- Machine Learning and Graph Representation Learning  
Topics in Large-scale Graph Learning, Optimization & Parallel Computing; Geometric Deep Learning (spatiotemporal analysis); Trustworthy ML (fairness, location privacy, robustness).

## PUBLICATIONS & MANUSCRIPTS

- Prasita Mukherjee, **Haoteng Yin**, Susheel Suresh, Tiark Rompf. OCTAL: Graph Representation Learning for LTL Model Checking. *arXiv preprint arXiv:2207.11649*
- Rongzhe Wei, **Haoteng Yin**, Junteng Jia, Austin R. Benson, Pan Li. Understanding Non-linearity in Graph Neural Networks from the Bayesian-Inference Perspective. *arXiv preprint arXiv:2207.11311*
- **Haoteng Yin**, Muhan Zhang, Yanbang Wang, Pan Li. Algorithm and System Co-design for Efficient Subgraph-based Graph Representation Learning. *Proceedings of the VLDB Endowment 15(11): 2788-2796*. DOI: 10.14778/3551793.3551831 (**VLDB'22**)
- **Haoteng Yin**, Yanbang Wang, Pan Li. Revisiting graph neural networks and distance encoding from a practical view. *The Fifth International Workshop on Deep Learning on Graphs (DLG-AAAI'21)*
- Bing Yu\*, **Haoteng Yin**\*, Zhanxing Zhu. ST-UNet: A Spatio-Temporal U-Net for Graph-structured Time Series Modeling. *arXiv preprint arXiv:1903.05631* (\* equal contribution)
- Bing Yu\*, **Haoteng Yin**\*, Zhanxing Zhu. Spatio-Temporal Graph Convolutional Networks: A Deep Learning Framework for Traffic Forecasting. *The 27th International Joint Conference on Artificial Intelligence (IJCAI'18)*. Main track. (\* equal contribution)
- **YIN Haoteng** and LIU Yang. Semantic analysis of spatial temporal trajectory in LBSNs. (in Chinese) *SCIENTIA SINICA Informationis 47(8), 1051-1065(2017)*. DOI: 10.1360/N112016-00310 (**Best Paper Award**)

## PROFESSIONAL SERVICE

- Graduate Teaching Assistant - CS242, CS38003, CS490-LDA, CS590-DE I&II Aug. 2019 - May. 2021
- Reviewer Dec. 2018 -
  - ACM Transactions on Spatial Algorithms and Systems
  - IEEE Transactions on Big Data, Cybernetics, and Intelligent Transportation Systems
  - NeurIPS'22, ICML'22, WWW'22, IEEE SPAWC'21 (TPC Member)
- Student Volunteer - The International Machine Learning Society Jul. 2018

## TECHNICAL SKILLS

- **Languages**: Chinese (native), English (professional working proficiency, OEPT certified)
- **Programming**: Proficient in Python/C (hybrid), PyTorch, Java, L<sup>A</sup>T<sub>E</sub>X, Git  
Experienced in C++, OpenMP, TensorFlow, Julia, R, MATLAB

Aug. 1, 2022

## RESEARCH EXPERIENCE

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- **Purdue University** West Lafayette, IN  
*Graduate Assistantship* Aug. 2019 - Current
  - **Efficient Subgraph-based Graph Representation Learning (SGRL)**  
Research Project (supported by GCoM@CS, Purdue), under the supervision of Prof. Pan Li.  
Design a novel framework SUREL for scalable SGRL by co-designing the learning algorithm and its system support. SUREL adopts walk-based decomposition of subgraphs and reuses the walks to form subgraphs, which substantially reduces the redundancy of subgraph extraction and supports parallel computation.
- **LIX, École Polytechnique & Inria** Palaiseau, France  
*Research Intern* Jun. 2018 - Sept. 2018
  - **Machine Learning Attacks to Location Privacy**  
Research Project (supported by team Comète, EP), under the supervision of Prof. Catuscia Palamidessi.  
Developed a model of adversary that uses machine learning algorithms to learn the trajectory pattern and the privacy mechanism of perturbed geospatial data gathered from location-based services, then performs privacy attacks on users' location.
- **Center for Data Science, Peking University** Beijing, P.R. China  
*Graduate Research Assistant* Jan. 2017 - Aug. 2019
  - **Traffic Prediction with Deep Spatial Temporal Graph Neural Networks**  
Research Project (supported by Deep Learning Lab, PKU), under the supervision of Prof. Zhanxing Zhu.  
Designed a GNN-based general deep learning framework (STGCN) to coherently model the topology of road networks and forecast future traffic conditions of the network through space-time streaming data in the mid-and-long term. It has been widely used as a SoTA baseline for traffic studies (cited over 1,490 times).
- **School of Computer Science and Technology, Shandong University** Ji'nan, P.R. China  
*Undergraduate Researcher* Jan. 2016 - May. 2017
  - **Hotspot Prediction Based on Temporal Trajectory and Social Attributes**  
Research Project (supported by Database Lab, SDU), under the supervision of Prof. Yang Liu.  
Proposed a location-vector embedding framework (Loc2vec) to predict geospatial hotspots in a certain area and explore their semantic meanings based on temporal trajectories that are collected from users' check-ins in location-based social networks.

## SELECTED HONORS & AWARDS

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- Graduate School Summer Research Grants, Purdue University Mar. 2021
- Academic Excellence Scholarship, Peking University (~8%) Oct. 2016, 2017, 2018
- The 35th International Conference on Machine Learning (ICML'18) Student Volunteer Award May. 2018
- The 2nd DiDi-IEEE Elite Forum Extraordinary Potential Award (2/39) May. 2018
- École Polytechnique International Student (Intern) Fellowship Mar. 2018
- The Most Technical Difficulty Award (Team Leader), Schlumberger HackPKU May. 2017
- Guanghua-Yintai PKU Social Innovation DO Camp Scholarship (€1,600) Jan. 2017
- The 5th National Conference of Social Media Processing the Best Paper Award (1/25) Oct. 2016
- The Best Ideas for Banking Business, AK BARS Bank Jul. 2016
- The Best Show Prize for the 3rd Taishan College Academic Forum May. 2016
- The 3rd Prize Excellent Student Scholarship, Shandong University Dec. 2013, 2014, 2015
- Taishan College International Study Scholarship (¥33,000) Jun. 2015
- The 2nd Prize for the 1st Taishan College Academic Forum Mar. 2014
- The Provincial 2nd Prize for the China Undergraduate Mathematical Contest in Modeling Dec. 2013
- The 2nd Prize Excellent Student Scholarship, Shandong University Dec. 2012